Pattern of Local Recurrence After I-125 Episcleral Brachytherapy for Uveal Melanoma in a Spanish Referral Ocular Oncology Unit

SARA SÁNCHEZ-TABERNERO, CIRO GARCÍA-ALVAREZ, M[A] FE MUÑOZ-MORENO, PATRICIA DIEZHANDINO, PILAR ALONSO-MARTÍNEZ, JESUS M. DE FRUTOS-BARAJA, FRANCISCO LÓPEZ-LARA, AND MARIA A. SAORNIL

• PURPOSE: To describe the time, frequency, and clinical characteristics of treatment failure after I-125 brachytherapy in patients with uveal melanoma treated and followed in a Spanish referral ocular oncology unit.

• DESIGN: Prospective, consecutive, interventional case series.

• METHODS: Patients diagnosed with uveal melanoma from 1995 to 2016 and treated with episcleral brachytherapy were included. Demographic data collection, ophthalmic evaluation, ultrasound scan, and systemic studies were performed at baseline, every 6 months thereafter for 5 years, and subsequently at annual intervals. Recurrence was defined as presence of tumor growth after treatment. Baseline analysis was performed by descriptive methods and survival by Kaplan-Meier curves.

• RESULTS: From 732 patients diagnosed with uveal melanoma, 311 were treated with brachytherapy. In the follow-up (mean 79 months, standard deviation = 55), 16 local tumor recurrences (5.1%) were detected. All relapsing patients had choroidal tumors and 15 presented with visual symptoms. All patients were treated with I-125 brachytherapy, and 2 received associated transpupillary thermotherapy. All the eyes were enucleated after recurrence. Kaplan-Meier analysis showed a mean time of recurrence of 3.7 years (standard deviation = 2.94 years, ranging from 1 to 12 years). Three patients had metastasis in the follow-up. Kaplan-Meier analysis showed worse survival for patients with recurrence.

• CONCLUSION: Local treatment failure was a relatively infrequent event after I-125 brachytherapy in our series. Recurrences appear not only early but also late in the follow-up. They do not have a distinctive clinical pattern and are associated with poorer survival. (Am J Ophthalmol 2017;180:39–45. © 2017 Elsevier Inc. All rights reserved.)

AJO.com Supplemental Material available at AJO.com.
Accepted for publication May 20, 2017.

From the Servicio de Oftalmología, Hospital Clínico Universitario de Valladolid (S.S.-T.); Unidad de Tumores Intraoculares del Adulto, Servicio de Oftalmología, Hospital Clínico Universitario de Valladolid (C.G.-A., M.A.S.); Unidad de Tumores Intraoculares del Adulto, Unidad de investigación, Hospital Clínico Universitario de Valladolid (M.F.M.-M.); and Unidad de Tumores Intraoculares del Adulto, Servicio de Oncología Radioterápica, Hospital Clínico Universitario de Valladolid (P.D., P.A.-M., J.M.F.-B., F.L.-L.), Valladolid, Spain.

Inquiries to Sara Sánchez-Tabernero, Servicio de Oftalmología, Hospital Clínico Universitario de Valladolid, Av Ramón y Cajal 12, 8B, 47003 Valladolid, Spain; e-mail: s.t.sara.g@gmail.com

UVEAL MELANOMA IS THE MOST COMMON PRIMARY intraocular malignant tumor in adults. It has an incidence of 4.3–10.9 cases per million people in a year.[1] Higher rates of uveal melanoma have been observed for male individuals and for light-skinned and for light-eyed individuals.[2,3] Cancer registries in Europe show a north-to-south decreasing incidence, from >8 per million in Norway and Denmark to <2 per million in southern Europe.[4]

Enucleation is still a treatment modality in use for uveal melanoma. It has, however, been largely replaced, when tumor size permits, by eye-conserving therapies. The most used option worldwide is episcleral plaque radiotherapy. This form of brachytherapy for uveal melanoma has demonstrated high rates of tumor control (89.7%) and eye preservation (87.5%) 5 years after treatment[5] and no significant difference in mortality when compared to enucleation by the Collaborative Ocular Melanoma Study for a 12-year follow-up.[6,7] The 2 most extended isotopes for plaque brachytherapy are iodine-125 and ruthenium-106. In a recent meta-analysis by Chang and McCannel, both isotopes were associated with a weighted mean local relapse rate of 9.6%.[8] Patients with local recurrence of the tumor might have an increased risk of metastatic disease and melanoma-specific death.[9,10]

The most used treatment modality for uveal melanoma patients at our ocular oncology unit is iodine-125 brachytherapy.[11] In a previous study with a 3-year average follow-up period we reported a low rate of treatment failure.[12] The purpose of this report is to document the incidence of local tumor recurrence after iodine-125 brachytherapy for uveal melanoma at a Spanish intraocular tumor referral unit. We also report how long after treatment the relapse occurred and we try to establish a clinical profile of patients who might be at risk of recurrence after treatment.

0002-9394/$36.00
http://dx.doi.org/10.1016/j.ajo.2017.05.019